

Manufacturing Operations Manual

Veritas Forge Industries Ltd

CHANAKYA Demonstration Dataset - synthetic data for a fictional company. Not real.

1. Safety Briefing and Readiness

Each shift begins with a 10-minute safety briefing and a machine readiness checklist. Changeovers follow the documented SOP and must be signed off by the line supervisor. Guards may never be bypassed.

1.1 Shift Schedule

Shift	Start	End
Shift 1	06:00	14:00
Shift 2	14:00	22:00
Shift 3	22:00	06:00

2. Changeover Procedure

Changeover steps: stop the machine, isolate energy, remove the previous tooling, clean the fixture, load the new programme, run first-article inspection and obtain supervisor sign-off before releasing production. Target changeover time is 45 minutes.

3. Production Planning

Weekly production plans are issued on Friday for the following week. Line capacity is planned at 85 percent of rated capacity to absorb breakdowns.

4. Machine A-01 (Line A) CNC Turning Centre

Machine A-01 is a CNC Turning Centre installed on Line A. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 250 running hours.

4.1 Specification A-01

Parameter	Value
Machine ID	A-01
Line	Line A
Type	CNC Turning Centre
Rated Capacity (units per shift)	185
PM Interval (hours)	250
Coolant Concentration (%)	6
Spindle Warm-up (minutes)	10

5. Machine A-02 (Line A) Vertical Machining Centre

Machine A-02 is a Vertical Machining Centre installed on Line A. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 300 running hours.

5.1 Specification A-02

Parameter	Value
Machine ID	A-02
Line	Line A
Type	Vertical Machining Centre
Rated Capacity (units per shift)	208
PM Interval (hours)	300
Coolant Concentration (%)	7
Spindle Warm-up (minutes)	15

6. Machine A-03 (Line A) Cylindrical Grinder

Machine A-03 is a Cylindrical Grinder installed on Line A. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 350 running hours.

6.1 Specification A-03

Parameter	Value
Machine ID	A-03
Line	Line A
Type	Cylindrical Grinder
Rated Capacity (units per shift)	231
PM Interval (hours)	350
Coolant Concentration (%)	8
Spindle Warm-up (minutes)	20

7. Machine A-04 (Line A) Horizontal Boring Machine

Machine A-04 is a Horizontal Boring Machine installed on Line A. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 400 running hours.

7.1 Specification A-04

Parameter	Value
Machine ID	A-04
Line	Line A
Type	Horizontal Boring Machine
Rated Capacity (units per shift)	194
PM Interval (hours)	400
Coolant Concentration (%)	6
Spindle Warm-up (minutes)	10

8. Machine A-05 (Line A) Gear Hobber

Machine A-05 is a Gear Hobber installed on Line A. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 450 running hours.

8.1 Specification A-05

Parameter	Value
Machine ID	A-05
Line	Line A
Type	Gear Hobber
Rated Capacity (units per shift)	217
PM Interval (hours)	450
Coolant Concentration (%)	7
Spindle Warm-up (minutes)	15

9. Machine A-06 (Line A) Heat Treatment Furnace

Machine A-06 is a Heat Treatment Furnace installed on Line A. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 500 running hours.

9.1 Specification A-06

Parameter	Value
Machine ID	A-06
Line	Line A
Type	Heat Treatment Furnace
Rated Capacity (units per shift)	180
PM Interval (hours)	500
Coolant Concentration (%)	8
Spindle Warm-up (minutes)	20

10. Machine A-07 (Line A) Wire EDM

Machine A-07 is a Wire EDM installed on Line A. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 550 running hours.

10.1 Specification A-07

Parameter	Value
Machine ID	A-07
Line	Line A
Type	Wire EDM
Rated Capacity (units per shift)	203
PM Interval (hours)	550
Coolant Concentration (%)	6
Spindle Warm-up (minutes)	10

11. Machine A-08 (Line A) Honing Machine

Machine A-08 is a Honing Machine installed on Line A. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 600 running hours.

11.1 Specification A-08

Parameter	Value
Machine ID	A-08
Line	Line A
Type	Honing Machine
Rated Capacity (units per shift)	226
PM Interval (hours)	600
Coolant Concentration (%)	7
Spindle Warm-up (minutes)	15

12. Machine B-01 (Line B) CNC Turning Centre

Machine B-01 is a CNC Turning Centre installed on Line B. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 250 running hours.

12.1 Specification B-01

Parameter	Value
Machine ID	B-01
Line	Line B
Type	CNC Turning Centre
Rated Capacity (units per shift)	186
PM Interval (hours)	250
Coolant Concentration (%)	6
Spindle Warm-up (minutes)	10

13. Machine B-02 (Line B) Vertical Machining Centre

Machine B-02 is a Vertical Machining Centre installed on Line B. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 300 running hours.

13.1 Specification B-02

Parameter	Value
Machine ID	B-02
Line	Line B
Type	Vertical Machining Centre
Rated Capacity (units per shift)	209
PM Interval (hours)	300
Coolant Concentration (%)	7
Spindle Warm-up (minutes)	15

14. Machine B-03 (Line B) Cylindrical Grinder

Machine B-03 is a Cylindrical Grinder installed on Line B. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 350 running hours.

14.1 Specification B-03

Parameter	Value
Machine ID	B-03
Line	Line B
Type	Cylindrical Grinder
Rated Capacity (units per shift)	232
PM Interval (hours)	350
Coolant Concentration (%)	8
Spindle Warm-up (minutes)	20

15. Machine B-04 (Line B) Horizontal Boring Machine

Machine B-04 is a Horizontal Boring Machine installed on Line B. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 400 running hours.

15.1 Specification B-04

Parameter	Value
Machine ID	B-04
Line	Line B
Type	Horizontal Boring Machine
Rated Capacity (units per shift)	195
PM Interval (hours)	400
Coolant Concentration (%)	6
Spindle Warm-up (minutes)	10

16. Machine B-05 (Line B) Gear Hobber

Machine B-05 is a Gear Hobber installed on Line B. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 450 running hours.

16.1 Specification B-05

Parameter	Value
Machine ID	B-05
Line	Line B
Type	Gear Hobber
Rated Capacity (units per shift)	218
PM Interval (hours)	450
Coolant Concentration (%)	7
Spindle Warm-up (minutes)	15

17. Machine B-06 (Line B) Heat Treatment Furnace

Machine B-06 is a Heat Treatment Furnace installed on Line B. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 500 running hours.

17.1 Specification B-06

Parameter	Value
Machine ID	B-06
Line	Line B
Type	Heat Treatment Furnace
Rated Capacity (units per shift)	181
PM Interval (hours)	500
Coolant Concentration (%)	8
Spindle Warm-up (minutes)	20

18. Machine B-07 (Line B) Wire EDM

Machine B-07 is a Wire EDM installed on Line B. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 550 running hours.

18.1 Specification B-07

Parameter	Value
Machine ID	B-07
Line	Line B
Type	Wire EDM
Rated Capacity (units per shift)	204
PM Interval (hours)	550
Coolant Concentration (%)	6
Spindle Warm-up (minutes)	10

19. Machine B-08 (Line B) Honing Machine

Machine B-08 is a Honing Machine installed on Line B. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 600 running hours.

19.1 Specification B-08

Parameter	Value
Machine ID	B-08
Line	Line B
Type	Honing Machine
Rated Capacity (units per shift)	227
PM Interval (hours)	600
Coolant Concentration (%)	7
Spindle Warm-up (minutes)	15

20. Machine C-01 (Line C) CNC Turning Centre

Machine C-01 is a CNC Turning Centre installed on Line C. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 250 running hours.

20.1 Specification C-01

Parameter	Value
Machine ID	C-01
Line	Line C
Type	CNC Turning Centre
Rated Capacity (units per shift)	187
PM Interval (hours)	250
Coolant Concentration (%)	6
Spindle Warm-up (minutes)	10

21. Machine C-02 (Line C) Vertical Machining Centre

Machine C-02 is a Vertical Machining Centre installed on Line C. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 300 running hours.

21.1 Specification C-02

Parameter	Value
Machine ID	C-02
Line	Line C
Type	Vertical Machining Centre
Rated Capacity (units per shift)	210
PM Interval (hours)	300
Coolant Concentration (%)	7
Spindle Warm-up (minutes)	15

22. Machine C-03 (Line C) Cylindrical Grinder

Machine C-03 is a Cylindrical Grinder installed on Line C. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 350 running hours.

22.1 Specification C-03

Parameter	Value
Machine ID	C-03
Line	Line C
Type	Cylindrical Grinder
Rated Capacity (units per shift)	233
PM Interval (hours)	350
Coolant Concentration (%)	8
Spindle Warm-up (minutes)	20

23. Machine C-04 (Line C) Horizontal Boring Machine

Machine C-04 is a Horizontal Boring Machine installed on Line C. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 400 running hours.

23.1 Specification C-04

Parameter	Value
Machine ID	C-04
Line	Line C
Type	Horizontal Boring Machine
Rated Capacity (units per shift)	196
PM Interval (hours)	400
Coolant Concentration (%)	6
Spindle Warm-up (minutes)	10

24. Machine C-05 (Line C) Gear Hobber

Machine C-05 is a Gear Hobber installed on Line C. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 450 running hours.

24.1 Specification C-05

Parameter	Value
Machine ID	C-05
Line	Line C
Type	Gear Hobber
Rated Capacity (units per shift)	219
PM Interval (hours)	450
Coolant Concentration (%)	7
Spindle Warm-up (minutes)	15

25. Machine C-06 (Line C) Heat Treatment Furnace

Machine C-06 is a Heat Treatment Furnace installed on Line C. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 500 running hours.

25.1 Specification C-06

Parameter	Value
Machine ID	C-06
Line	Line C
Type	Heat Treatment Furnace
Rated Capacity (units per shift)	182
PM Interval (hours)	500
Coolant Concentration (%)	8
Spindle Warm-up (minutes)	20

26. Machine C-07 (Line C) Wire EDM

Machine C-07 is a Wire EDM installed on Line C. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 550 running hours.

26.1 Specification C-07

Parameter	Value
Machine ID	C-07
Line	Line C
Type	Wire EDM
Rated Capacity (units per shift)	205
PM Interval (hours)	550
Coolant Concentration (%)	6
Spindle Warm-up (minutes)	10

27. Machine C-08 (Line C) Honing Machine

Machine C-08 is a Honing Machine installed on Line C. Operators must complete the readiness checklist at the start of every shift and record results in the shift logbook. Preventive maintenance is due every 600 running hours.

27.1 Specification C-08

Parameter	Value
Machine ID	C-08
Line	Line C
Type	Honing Machine
Rated Capacity (units per shift)	228
PM Interval (hours)	600
Coolant Concentration (%)	7
Spindle Warm-up (minutes)	15